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# DS
# Technical Case

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# Optimize coupon offers to convert casual customers and improve customer lifetime value

## Causal Models and Short-Term Proxies unlock a long-term customer engagement strategy

(Methodology)

- **From Correlation to Causation**: Shifting from classic ML (which only identifies associations) to a causal framework that captures true cause-and-effect.
- **The Limitation of Traditional Test Analysis**: Since we have many test parameters, standard methods like ANOVA fail due to the impossibility of creating clean comparison groups.
- **The Causal Solution:** Causal models learn the impact of each parameter, enabling data-driven optimization despite high complexity.

(Time window)

- **Short-Term Proxies**: Leveraging a 30-day window and short-term metrics to measure initial impact across multiple offers.
- **Long-Term Validation**: Continuously testing if these early metrics reliably predict our long-term goals as more data becomes available.

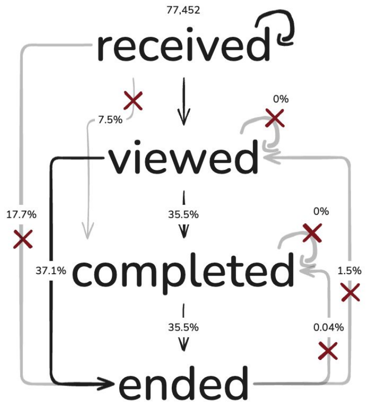
# Transforming Event Data to Offer-Level for Causal Modeling

## Data Transformations

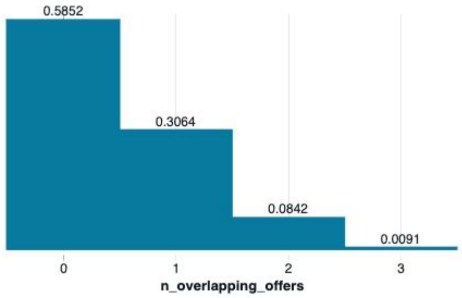
- **FIFO**: First-In-First-Out logic to link transactions events to a specific offer
  This selects 99.5% of the initial offer count (77k)
- **Non-compliance**: Remove non-viewed offers (72.6%)
- **More than one treatment**: Remove offers that are overlapping (58.5%)
- **Target Censorship**: Remove offers with fewer than 4 days between the next offer or last day (33.6%)

After applying all filters, 20,901 out of 77,452 offers were selected (27%) to create this model

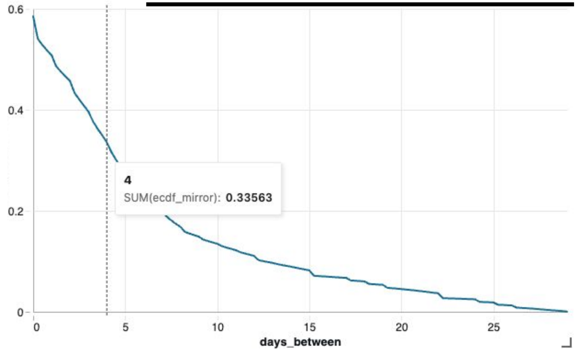
Offer States




Percentage of overlapping offers




P(Days Between Offers > x)



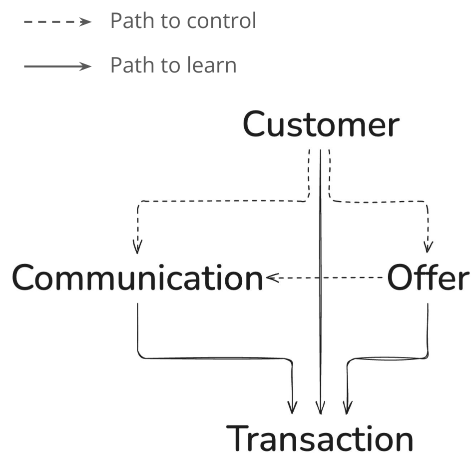
# Causal Modeling Pipeline & Technical Definitions

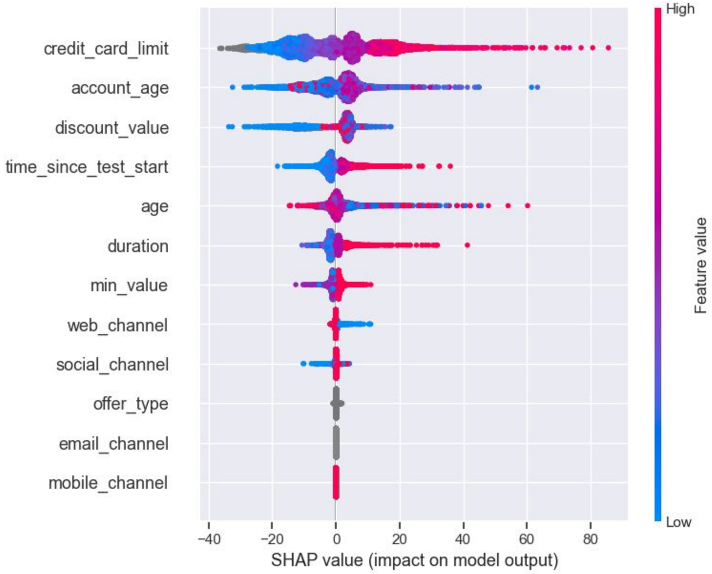
To control confounders, we used the backdoor criterion, adding customer and communication data to the model.

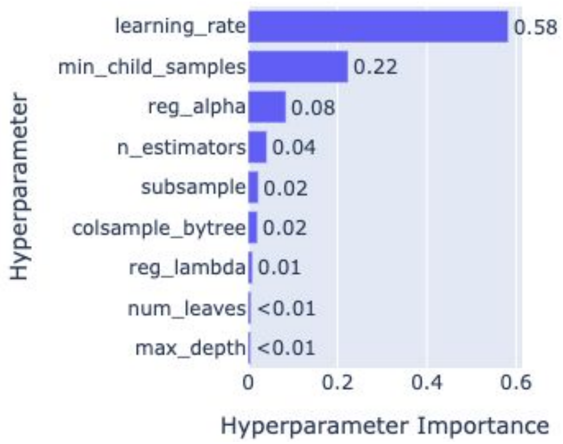
We developed two models: 1) customer information only, and 2) all information. Gender was excluded for fairness.

Model Stack: We used an LGBM Regressor, Optuna for hyperparameter optimization, and SHAP for feature impact analysis

Structural Causal Model (DAG)



# Financial Results

**Maximizes ROI early**: The green curve rises steeply, proving we can capture roughly 70% of our total business value by targeting only the top 50% of the population.

**Proves Causal Advantage**: Our final_model consistently outperforms the traditional, non-causal approach (orange line).